

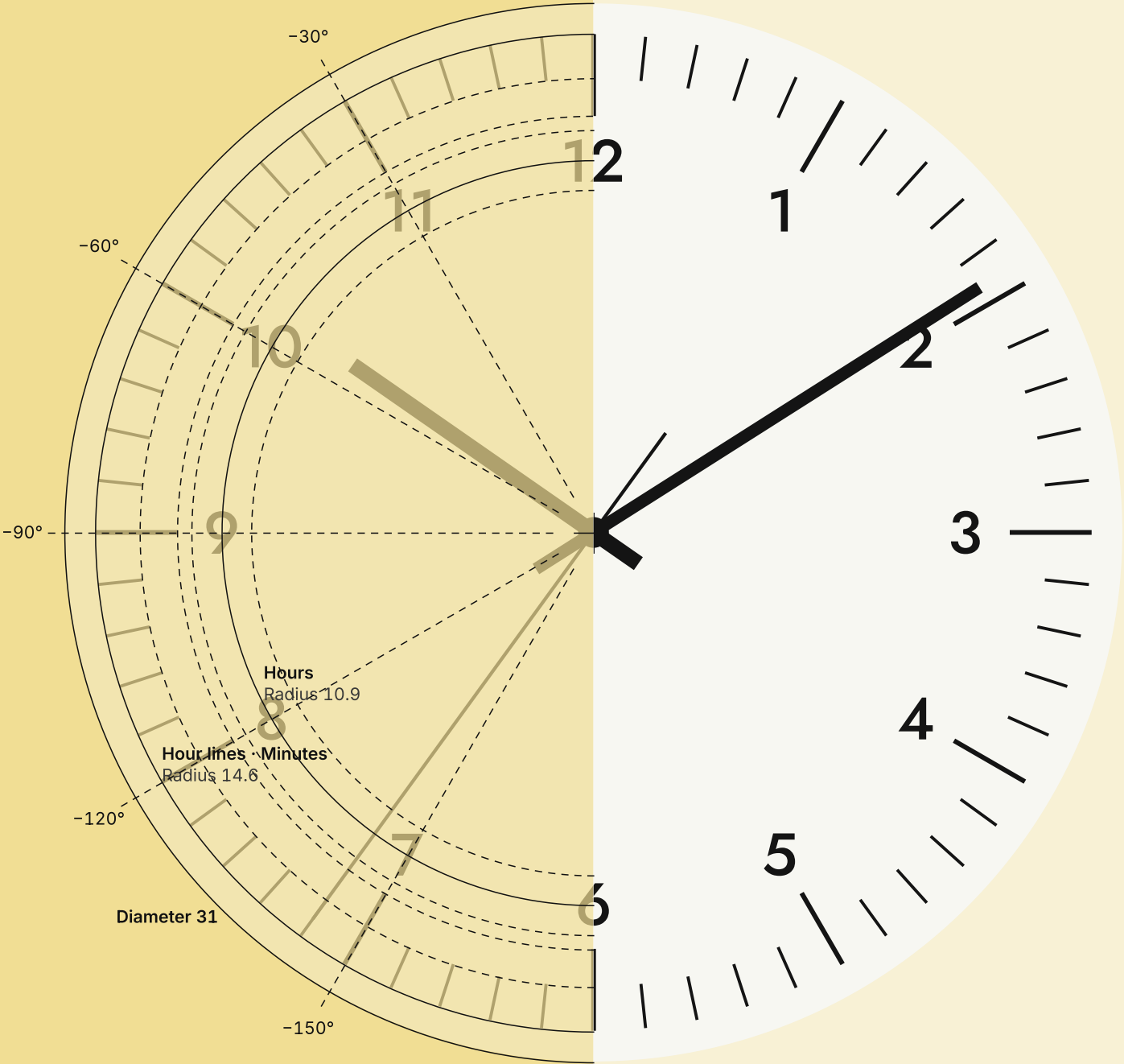
Dial Tool

Guidebook

beta 2.3 · 4 Oct 2026

A dial is a short list of rings. In the middle, the list: every radius and angle the settings name. On the right, what it prints.

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One Python file that runs inside DrawBot, a free drawing app for macOS.
Everything is in millimetres.



Install

Three steps, one file.

- 1 **DrawBot 3.132** or newer, from drawbot.com.
- 2 **dial.py** from github.com/duztman/dial-tool.
Branch main is the released version; dev is the newest work.
- 3 Open dial.py in DrawBot and press **⌘R**. The window opens.



Speed

One optional package.

Knockouts and hollow shapes are cut by combining outlines. DrawBot's own routine is slow; skia-pathops is 2 to 14 times faster.

In DrawBot: Python → Install Python Packages → **skia-pathops** → Go. The Export log then opens with "shape combining: skia-pathops".



Memory

Nothing to save.

Settings are stored as you work. Press **⌘R** again, or quit DrawBot: the last session reopens as you left it. Export → Save... writes a named copy; Load... opens one. The nine dials in this book come as such files.



Unit

mm

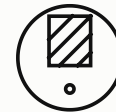
Every length is a millimetre, drawn at true size. Radii are measured from the dial's centre.



Angle

0° = 12

Clock angles, growing clockwise. 90° is 3 o'clock, 180° is 6, -90° is 9.



Shape

Fills

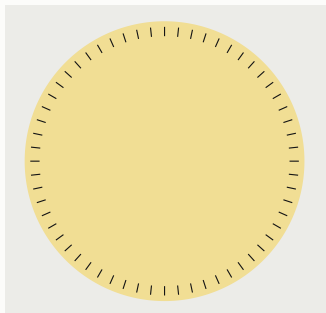
Everything exported is a filled outline, never a stroke. A 0.12 mm tick is 0.12 mm in Illustrator.



Order

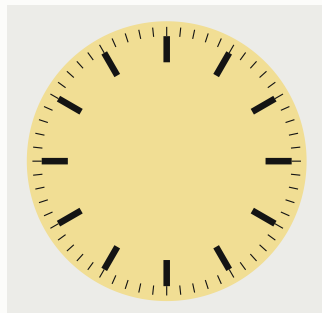
Top first

The ring list is the stacking order. The top ring is drawn last, in front of the others.



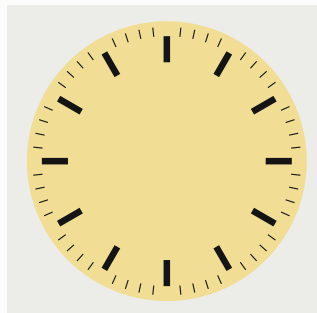
1 · ticks

Add ring... → ticks · Count 60



2 · markers

added above the ticks



3 · give way

ticks skip where markers sit



4 · numerals, hands

Every 3 · Radius 8 · Size 2.4



Watch

It runs only inside DrawBot. The window, the fonts and the export all come from DrawBot's own Python.



Watch

The canvas preview needs macOS 14 or newer. Older systems get the PDF preview, and the log says why.

The window

Settings on the left, the dial on the right, time underneath. Overleaf it is redrawn whole, in grey, from the tool's own measurements.

①

Sections

Dial, Rings, Hands, Date, Export. One is shown at a time.

②

Ring list

Top ring in front. The checkbox hides a ring; double-click a name to change it. The selected ring is the one the settings edit.

③

Ring tools

Add ring... puts a new ring above the selected one. Duplicate, Delete, move up, move down.

④

The ring's settings

One scrolling list. Rows that don't apply to the ring's kind are hidden, not greyed out.

⑤

Preview

Follows every change. Pinch or ⌘-scroll zooms about the cursor; two fingers pan. With Guides on, the selected ring's edges are orange.

⑥

Time

Drag, or type 10:09:36. Now sets the clock's time; Play runs the hands.

⑦

Guides, Canvas · PDF, Fit

Guides are never exported. Canvas is the fast preview; PDF is DrawBot's own viewer, the reference. Fit shows the whole page.

⑧

Status

Ring, font, diameter; then how long the geometry (build) and the drawing (frame) took.



Number

Slider, field and stepper show one value.

Length

Width · Ø

Drag for a rough value, type for an exact one, step for the last hundredth. Typing past a slider's end extends the slider.



Typing

Fields do arithmetic.

1.2	sets 1.2
+0.5	adds 0.5
x0.5	halves
/3	a third

Useful when a drawing says “a third of the way in” and the field says 13.4.



Choice

Popups, switches, checkboxes.

Kind

Shape ☐ bar ☐ wedge ☐ dot ☐ file

Style ☐ same ☐ double ☐ triangle ☐ none

A choice can change which rows are shown: pick dot and Length disappears, because a dot has none.

Dial Rings Hands Date Export

RingKind

Hoursnumerals

24 hoursnumerals

Hour markersmarkers

Minutesticks

Add ring...

DuplicateDelete

UpDown

Kindticks

Colour

Outer radius14.40

Positions

Count60

Start °0.0

Span °360

Every1

Offset0

Skip those instead

Give way to rings above

Skip wheremarkers above

Knock outnothing

Look

Shapebarwedge-dot-file

Length1.00

Width · Ø0.16

Round outer0.00

Round inner0.00

Dial Tool · beta 2.3

5

67

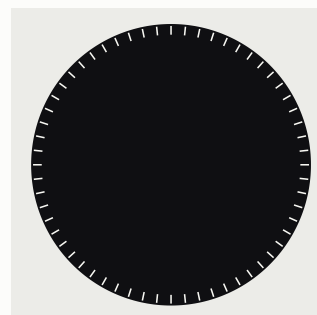
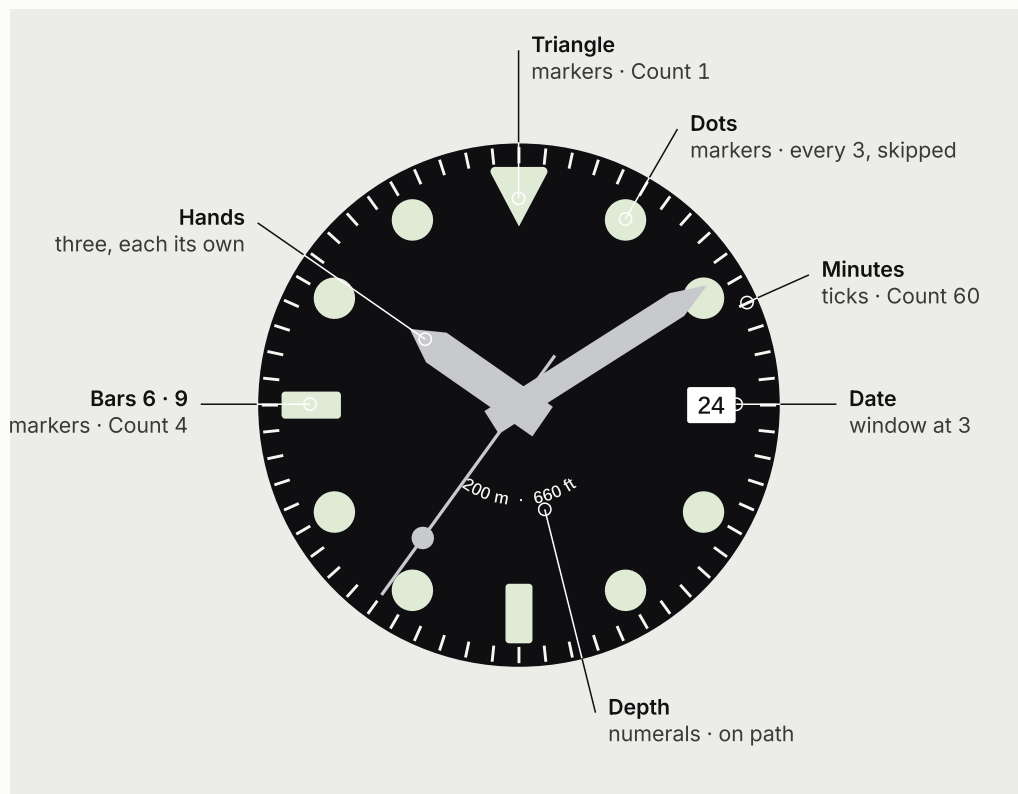
8

Minutes · Ø 30.0 mm · canvas · build 0 ms · frame 1 ms

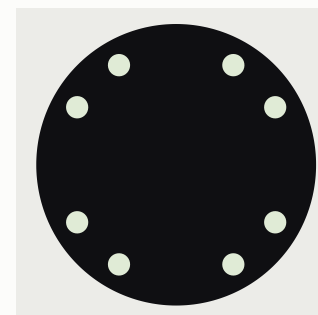
02

Rings

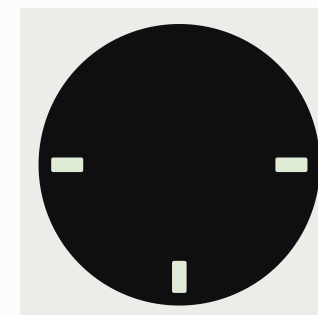
A ring repeats one thing round the centre, at its positions. A dial is a short list of rings, stacked.



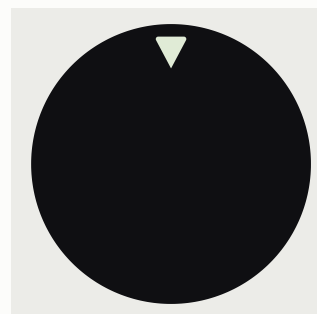
Minutes
ticks · Count 60



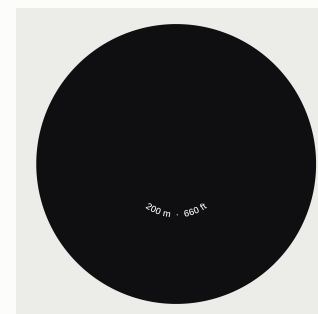
Dots
markers · Every 3, skipped



Bars 6 · 9
markers · Count 4 · 12: none



Triangle
markers · Count 1 · wedge



Depth
numerals · Custom · on path



The dial
with hands and the date

The diver's ring list

Ring	Kind
☒ Triangle	markers
☒ Bars 6 · 9	markers
☒ Dots	markers
☒ Depth	numerals
☒ Minutes	ticks



Watch

Order works twice. It is the stacking order, and a ring only gives way to rings **above** it. For ticks to stop at the markers, the markers go higher in the list.



The idea

One dial, five rings.

The diver of page 16, taken apart. Nothing is special-cased: the triangle is a markers ring with a count of one, the depth rating a numerals ring with one label.



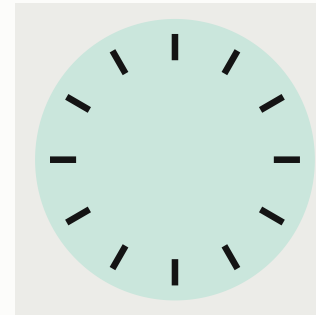
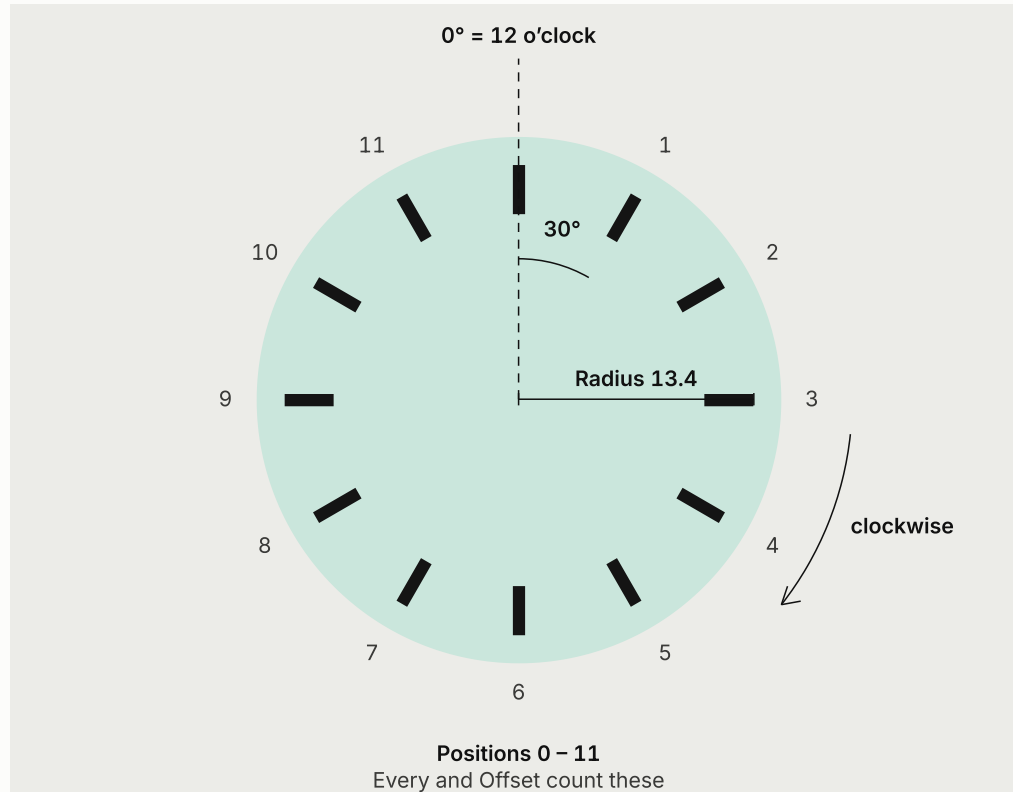
Kind

Four kinds.

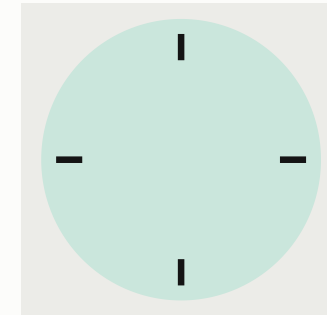
Ticks and markers repeat a shape, numerals a label; a band is colour between two radii. A ring can change kind at any time: it carries every setting with it.

Positions

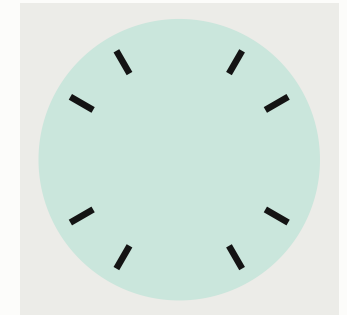
Where a ring puts things. Six numbers describe almost any track, scale or set of hours.



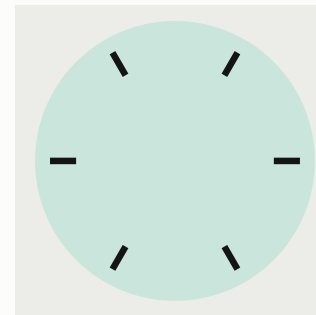
Count 12
Start 0° · Span 360°



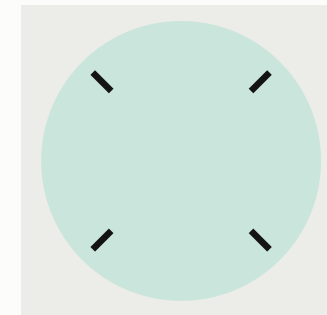
Every 3
the cardinals



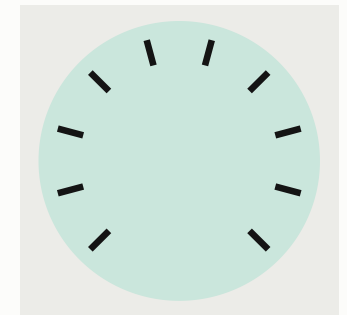
Every 3 · skip those instead
the other eight



Every 2 · Offset 1
the odd hours



Count 4 · Start 45°
a quarter turn off



Count 10 · Span 270°
Start -135° · both ends marked

The arc, sixth picture

Count

Start °

Span °

The cardinals, second picture

Every

Offset



Watch
Radii are absolute. Changing the dial's Diameter moves no ring and no hand. Set the diameter first, then each Radius, the way a movement's drawing would.



Count · Start · Span
An arc marks both ends.

Span 360° spaces the count evenly. Anything less puts the first and the last on the ends: Count 10 over 270° runs 0 to 9.

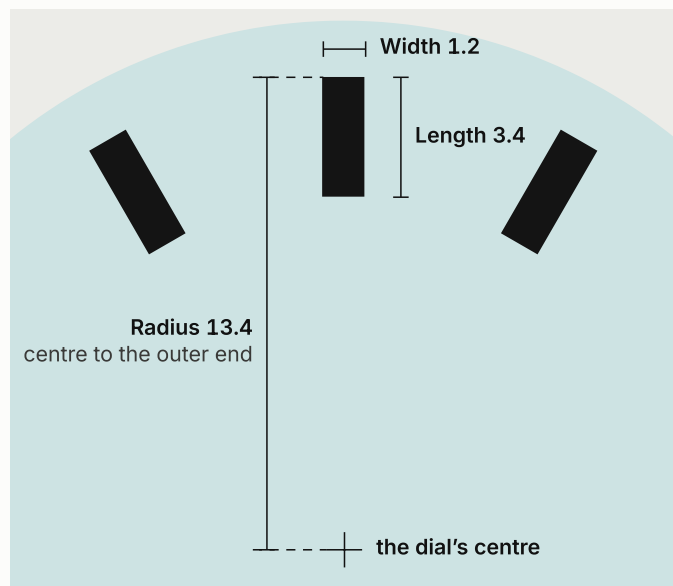


Every · Offset
Use some of them.

Every 3 keeps the cardinals; Offset shifts which ones. Skip those instead keeps the other eight: two rings can share one circle.

Marks

Ticks and markers share one set of shapes. A markers ring adds a first position that can differ from the rest.



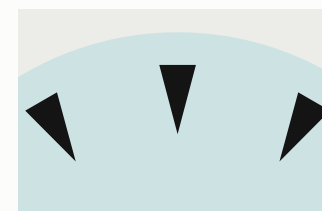
bar
Length 3.4 · Width 1.2



bar, fine
Count 60 · 1.6 × 0.12



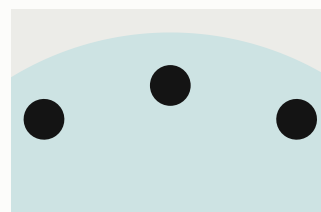
wedge
Inner width × 0.35



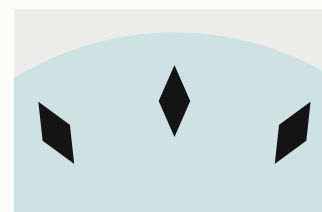
wedge
Inner width × 0



wedge
Inner width × 1.8



dot
Ø 2.0



file
your own SVG · PDF · AI



rounded
Round outer 0.6 · inner 0.2

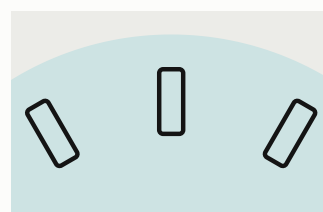
The mark above

Shape ☐ bar ☐ wedge ☐ dot ☐ file

Outer radius 13.40

Length 3.40

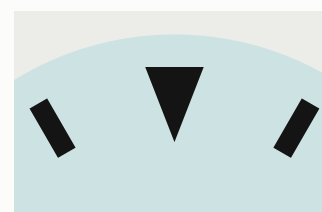
Width · Ø 1.20



hollow
Hollow wall 0.22



12: double
Gap 0.4



12: triangle
Scale × 1.2



12: none



Watch

A fillet can't exceed half the side it rounds, and a Hollow wall of half the width or more leaves the mark solid. Neither warns: the shape just stops changing.



Wedge and corners

One number tapers; two round.

Inner width × is the inner end as a share of the outer: 0 a triangle, 1 a bar, more widens inward. Round outer and Round inner are true fillets, in mm.



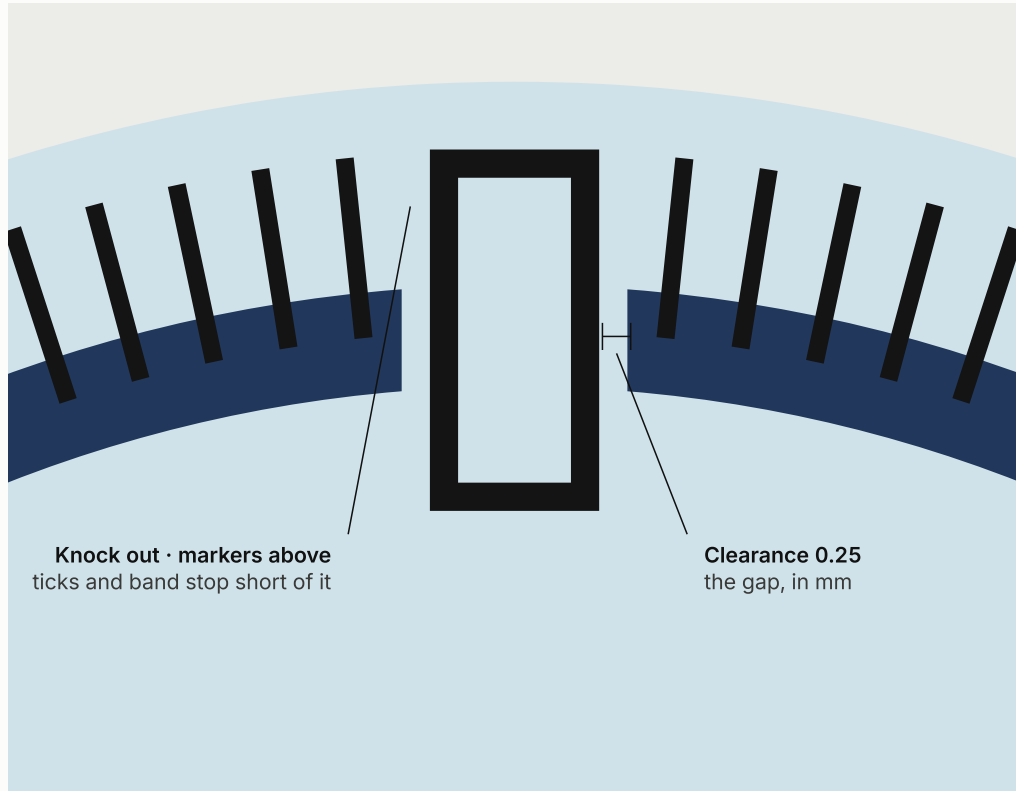
12 o'clock

The first position can differ.

Markers draw their first position the same, double (with a Gap), as a triangle, or not at all. Scale × sizes it. Ticks have no such position.

Give way

How a ring reacts to the rings above it. One mechanism covers every pair of rings.



The ticks above

Skip where

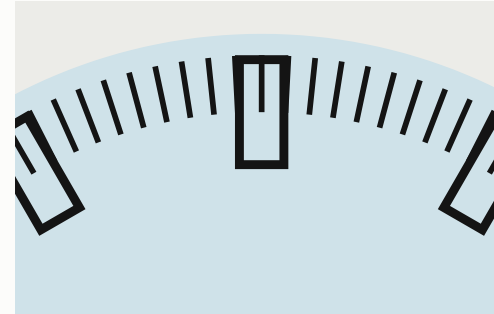
Knock out

Clearance

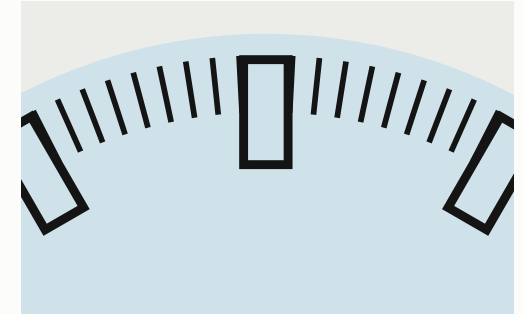


Watch

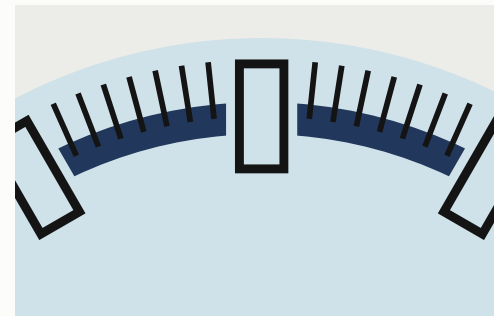
Skip compares angles to a hundredth of a degree. Two rings skip each other only where their positions truly coincide: same Start, counts that divide. Otherwise use Knock out, which looks at shapes, not angles.



stack
Skip where: nothing · Knock out: nothing



skip where
markers above



knock out
markers above · Clearance 0.25



knock out
markers & numerals above



Skip where

Leave out what is taken.

The ring drops its own positions wherever the chosen rings above already have a mark: minute ticks that stop for the hour markers.



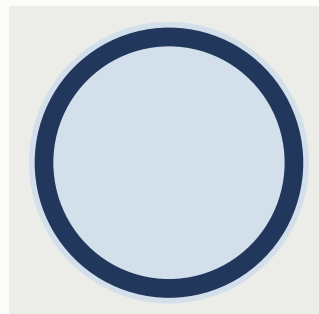
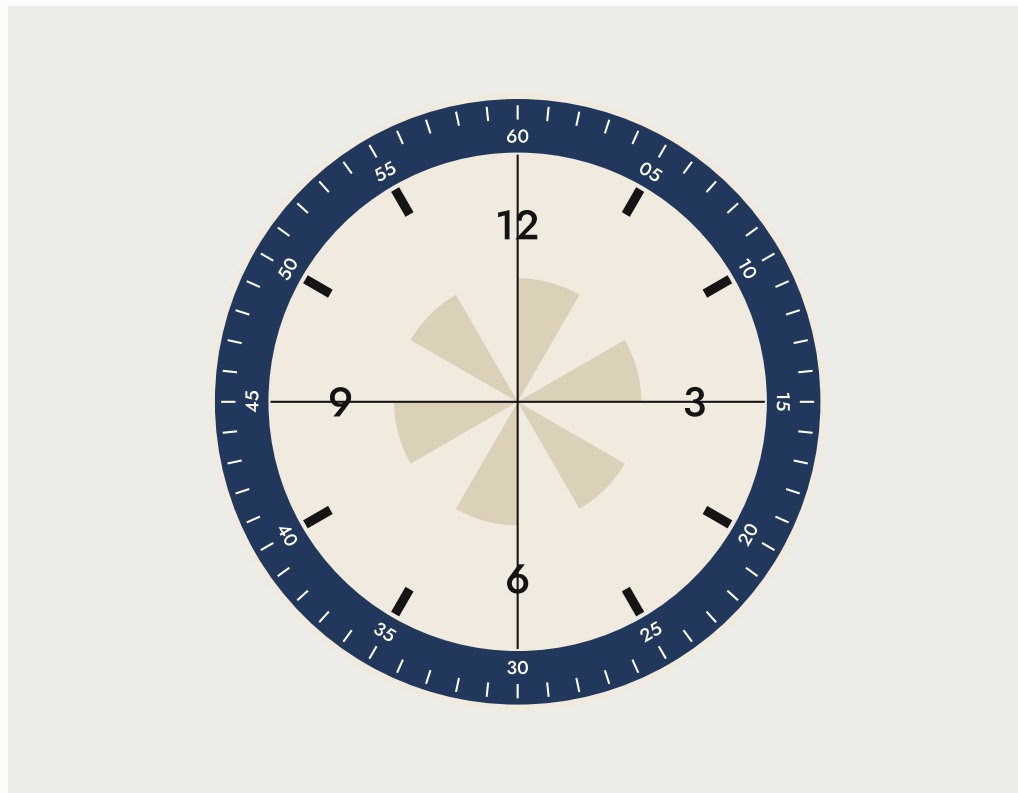
Knock out

Cut a gap round it.

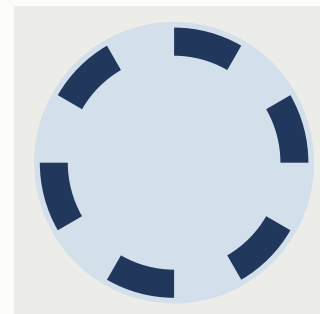
The ring is cut back by Clearance mm round the shapes above: markers, markers and numerals, or everything. The cut follows the shape, serif and all.

Bands

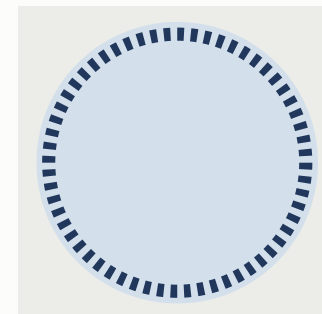
Flat colour between Radius and Radius – Band width: rings, rails, zones and sectors.



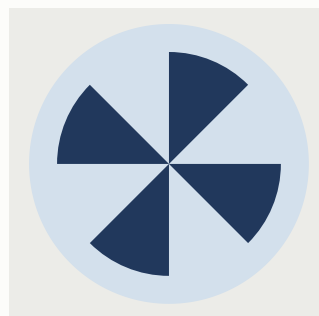
ring
Count 1 · Band width 2.0



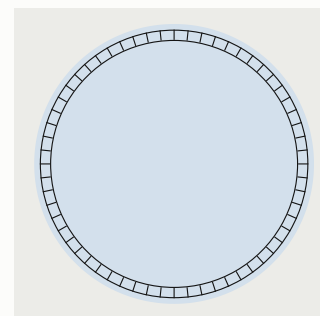
sectors
Count 12 · Every 2



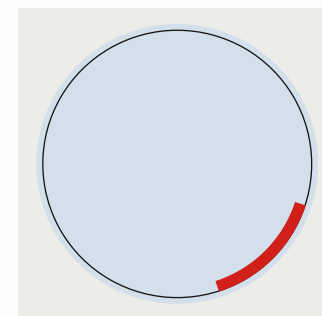
fill
Count 60 · Fill 50 %



pies
Band width = radius



rails
two bands, Band width 0.12



arc
Start 108° · Span 54°



Rails

A hairline is a band too.

Band width 0.12 draws a circle line.
Two of them and a ticks ring make a railroad track: page 19.



Watch

Colour is flat, one per ring. A chapter ring with white print is three rings: the band, then white ticks and numerals above it, as in the sector dial here and on page 22.



Segments

Positions cut a band.

Count 1 is a full ring. More positions cut it into segments that tile the span; Fill % is how much of its step each one covers.

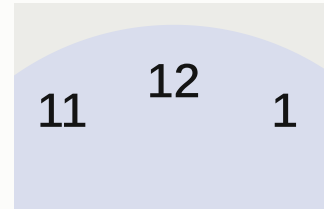
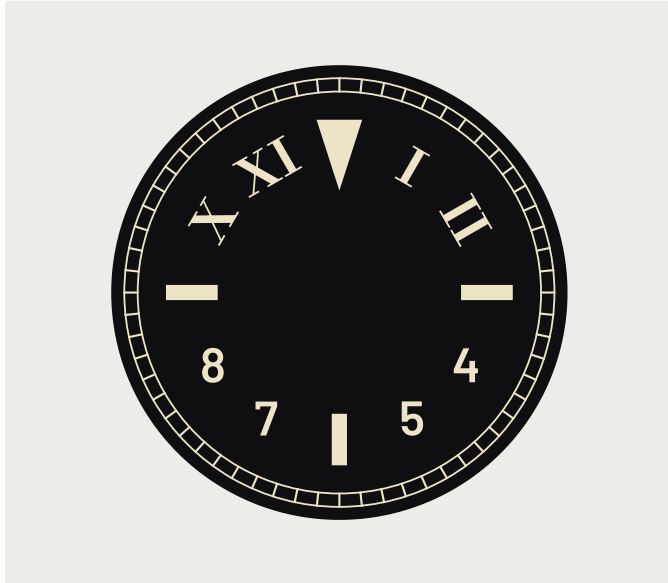


The sectors, second picture

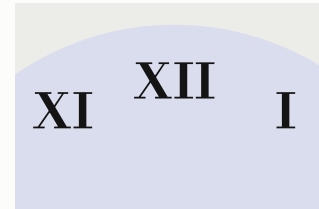
Outer radius	14.40
Band width	3.00
Count	12
Every	2

Numerals

One label per position: from a built-in set, a count, or a list of your own.



Arabic



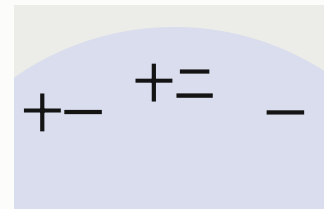
Roman · III



Roman · IV



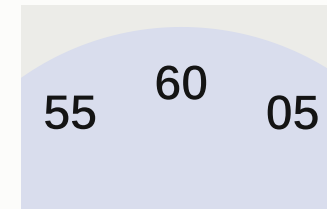
Eastern Arabic



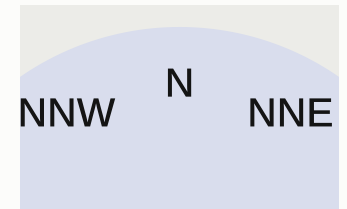
Chinese



Words
on path



Numbers
First 5 · Step 5 · Digits 2



Custom
N, NNE, , , E, ...

12

Numbers

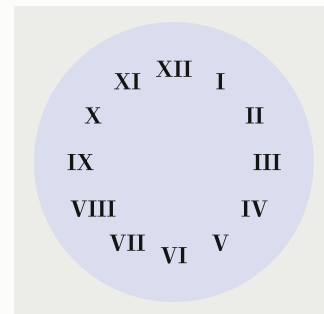
Labels that count.

Labels

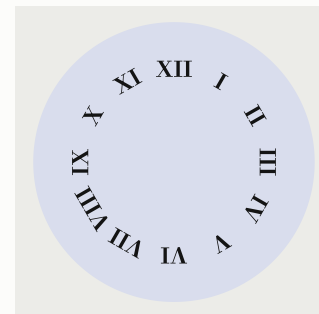
First number

Step

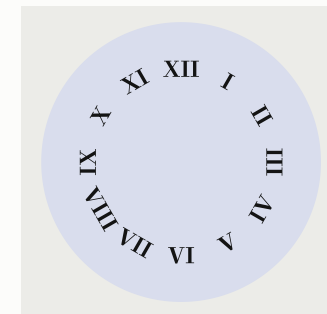
Digits



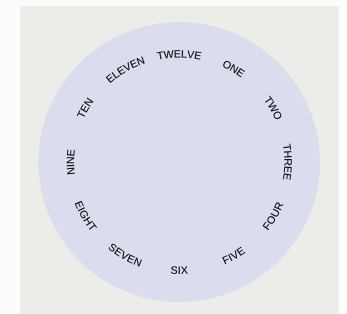
upright
always level



radial
turns with the circle



radial, auto-flip
never upside down



on path
letters follow the circle



Watch

A Custom list follows positions, not hours.
With Start 30° its first entry sits at 1 o'clock.



Placement

Four ways to sit.

On path sets each letter on the circle; below 3 and 9 it flips to read left to right.



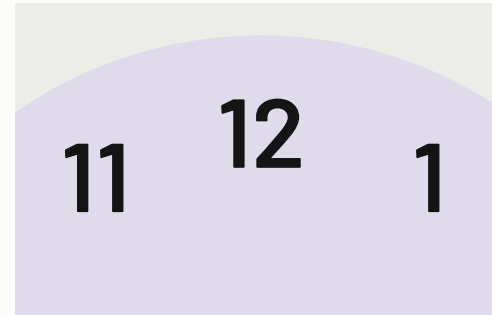
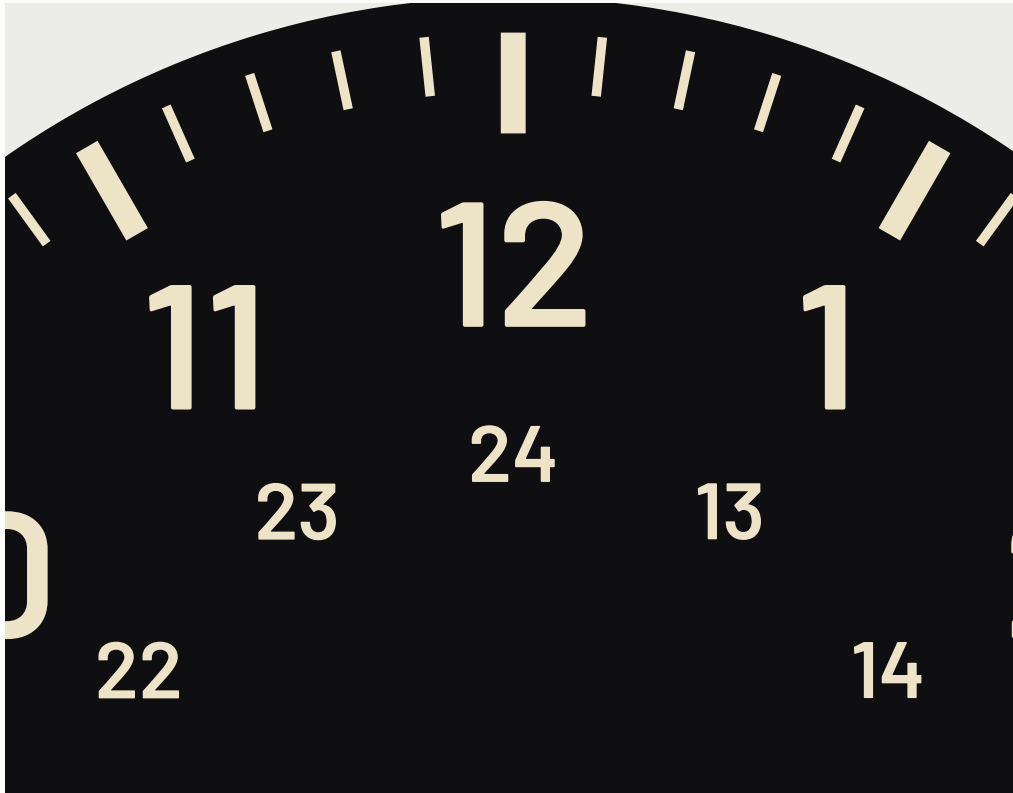
Custom

A list, comma by comma.

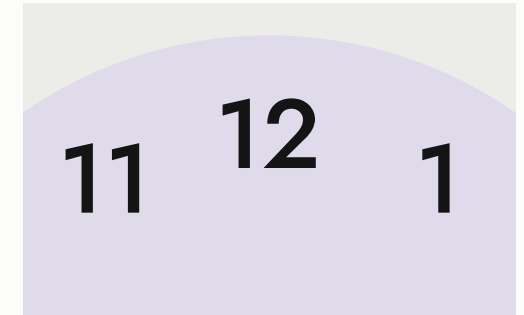
An empty entry leaves its position empty:
California, page 19, is two such lists.

Type and nudge

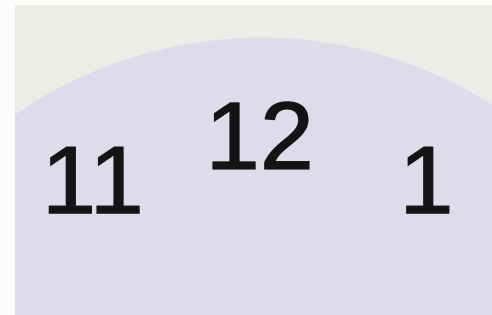
Each numerals ring has its own type, from any font on the Mac. Each numeral can then be corrected by eye.



DIN Alternate Bold



Futura Medium



Helvetica Neue Medium



Bodoni 72 Book



Nudge

One numeral at a time.

Radius ± in or out, mm

Angle ±° round the dial

Rotate ° about its centre

Size × larger, smaller



Watch

Two-digit numerals look heavier than single ones. Size × 0.92 on 10, 11 and 12 evens them; Radius -0.2 pulls a wide 12 inward. Nudges belong to the position, not the label: change Count and they stay where they were.



Type

What reaches the outlines.

Family · Style

read from macOS; there is no fonts folder

Instance · Axes

a variable font's named instances, and a slider per axis

Features

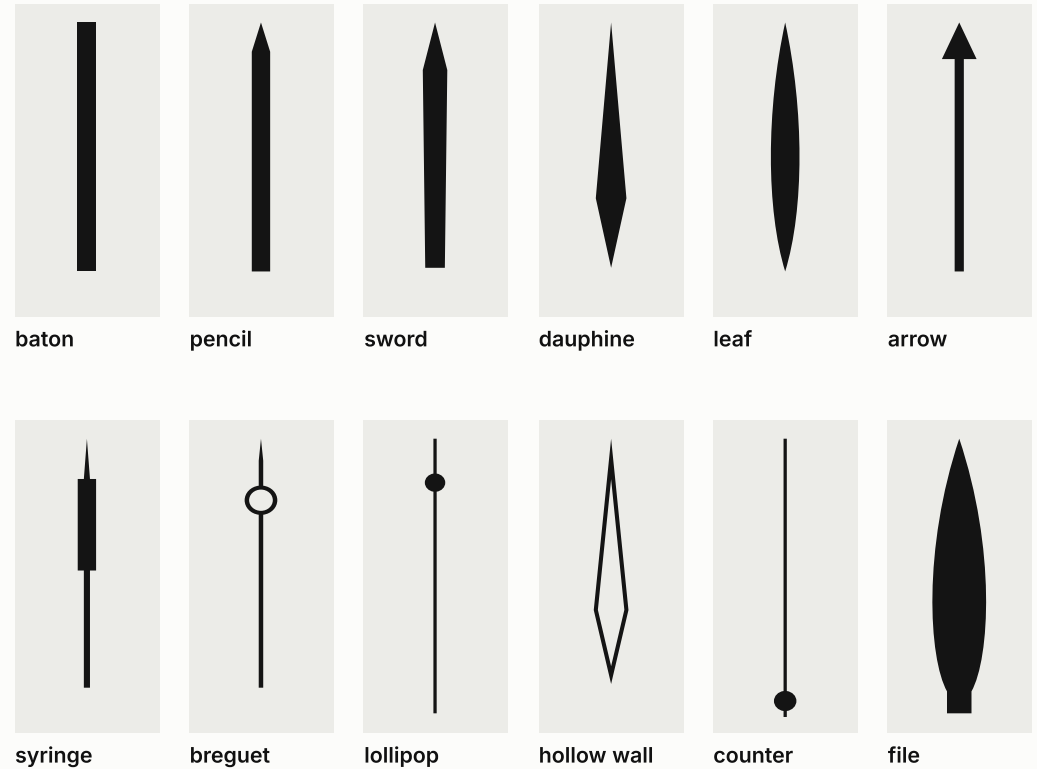
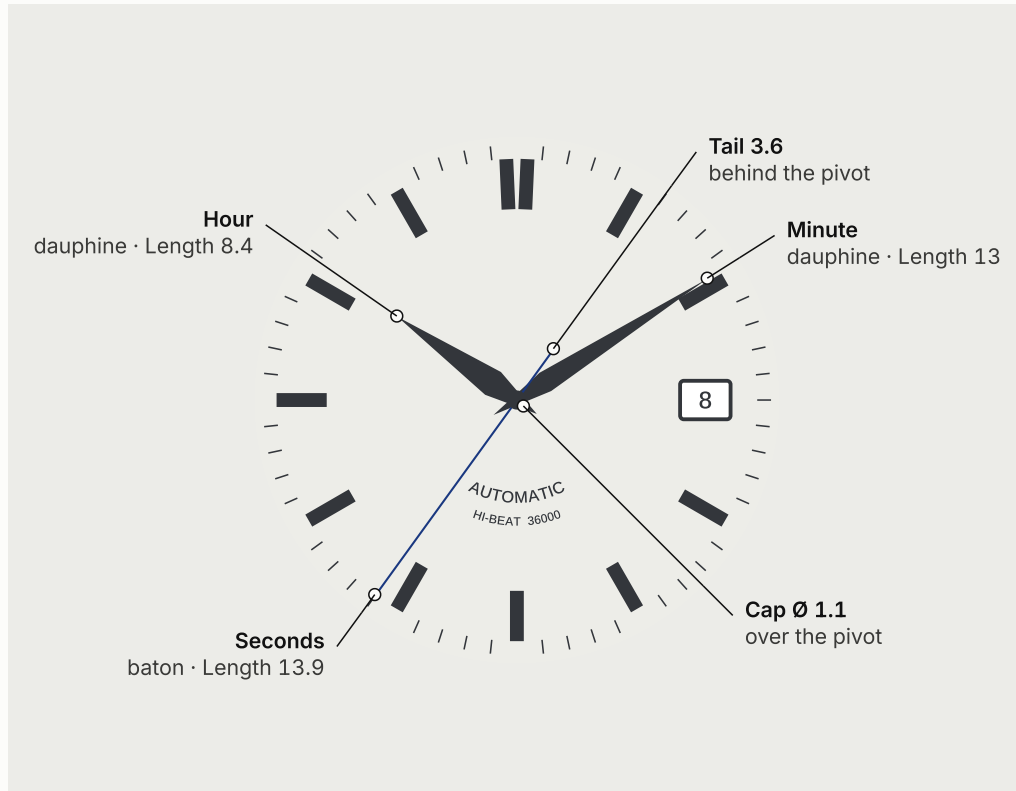
OpenType features, under the designer's own names

Language · Tracking

local forms, such as Turkish İ i · 1/1000 em, as in Illustrator

Hands

Hour, minute and seconds each have their own shape, size and colour. Lengths are millimetres from the pivot.



Length · Tail

Measured from the pivot.

Length runs from the pivot to the tip, Tail the other way. Counterweight Ø adds a disc to the tail; Hollow wall turns any hand into an outline.



Watch

Lengths don't follow the dial. A 13 mm minute hand on a 24 mm dial overhangs it by a millimetre, and nothing stops you.



The diver's minute hand

Shape	pencil	
Length	<input type="range"/>	12.40
Width	<input type="range"/>	1.50
Tail	<input type="range"/>	1.80
Tip length	<input type="range"/>	2.00



Motion

Four beats.

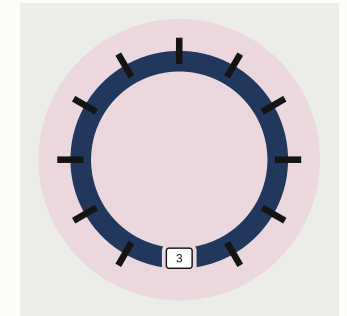
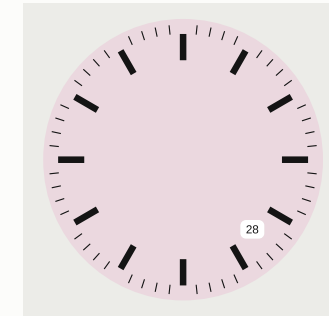
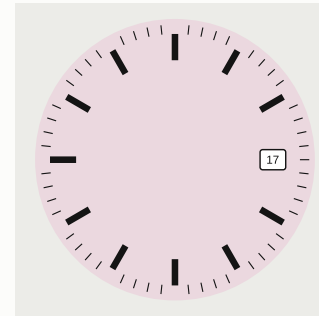
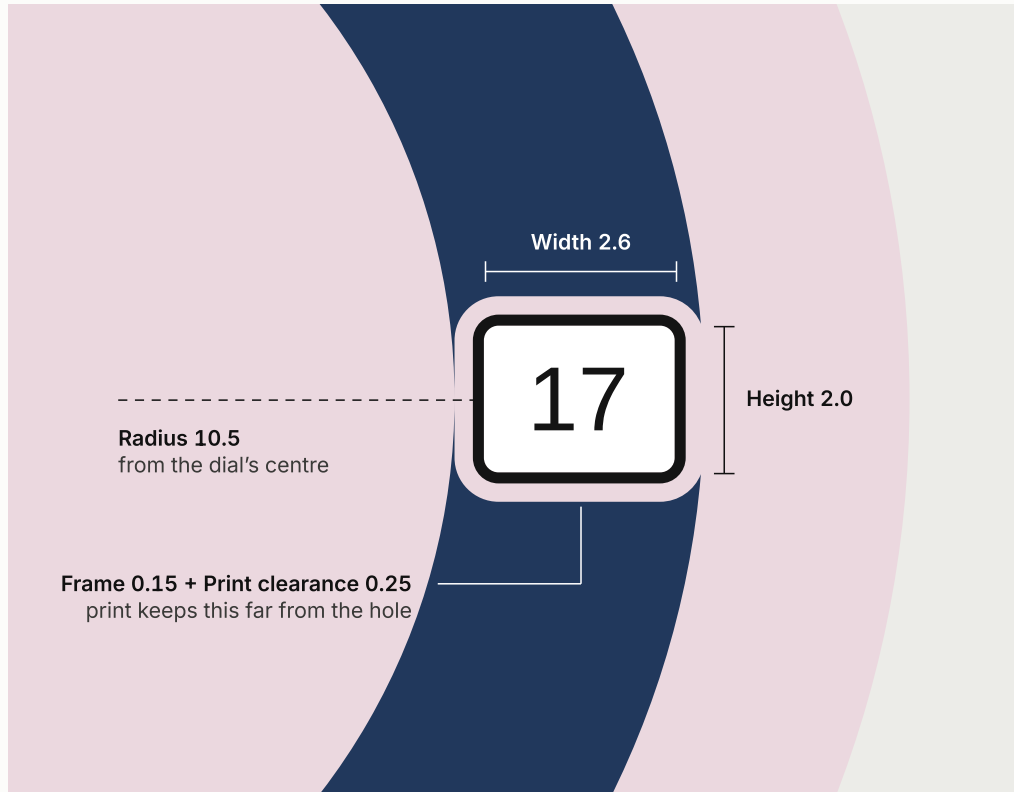
sweep smooth

quartz 1 step a second

6 · 8 beats/s 21,600 · 28,800 vph

Date window

A hole in the plate that shows the date disc, with an optional printed frame.
Print keeps its distance.



Clearance

Print keeps away.

Every ring is cut back by the frame plus Print clearance. Markers and numerals leave the date's position empty altogether.



Type

The top numerals ring's font.

Shows is the day drawn on the disc; Text % of height sizes it. 28 is the widest day: check with it.

The window above

Position

Radius

Width

Height

Corner

Frame

Print clearance



Watch

The frame and the day share one Print colour, and the disc is always white. On a dark plate a light frame would mean a light day on white: set Frame to 0 there, as the diver does.



Production

A cut line.

In a production mask the aperture is an outline to cut or drill to.



Position

Five places.

3, 4:30, 6, 9 or 12; Radius sets how far out.

Classics

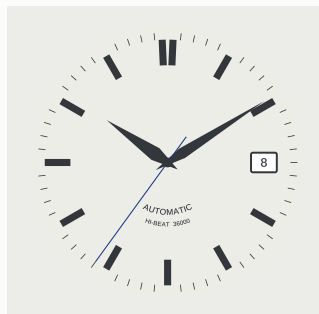
Nine dials after well-known types. Each page prints the settings that drew it.



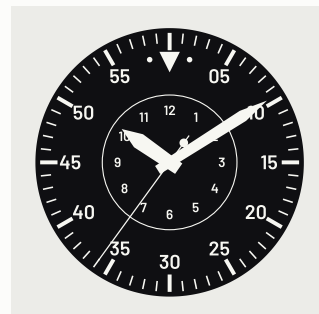
Field watch
page 15



Diver
page 16



Dress watch
page 17



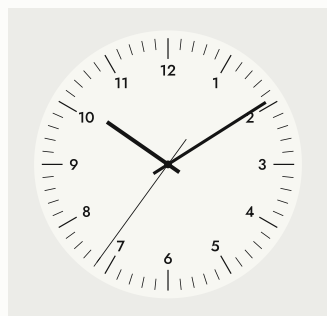
Pilot, type B
page 18



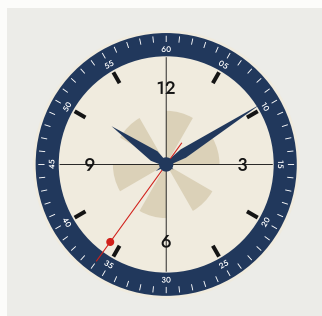
California
page 19



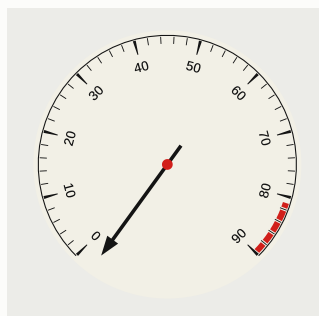
Station clock
page 20



Bauhaus
page 21



Sector dial
page 22



Gauge
page 23



Presets Load them.

Each dial is a settings file in *docs/guide/presets*. Export → Settings → Load... opens it, ready to change.



Reading a page Rings top first.

Only what differs from a newly added ring is printed; everything else is at its starting value.



How they were drawn By eye, not by caliper.

Proportions were judged against pictures of the type, not measured from a watch. Treat each as a starting point: load it, then move it toward yours.



Watch

Fonts are named as on macOS: DIN Alternate, Futura, Helvetica Neue, Bodoni 72. These pages were drawn with similar open fonts, so numerals differ slightly from what the presets draw on a Mac.



Watch

Load... replaces the whole dial: rings, hands and date. Save your own first, with Export → Settings → Save...



12.1

Field watch

After the A-11 of the 1940s and its descendants: 12 and 24 hours, a plain minute track, nothing else.



Look for
The 24-hour ring is a second numerals ring with a Custom list. The minute ticks **skip** where the hour marks sit, so the two never overprint.

Settings that differ from a new ring

Hours numerals	Radius 10.5 · Size 3 · Font DIN Alternate Bold ☐
24 hours numerals	Radius 7.3 · Labels Custom · Custom 24, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 · Size 1.35 · Font DIN Alternate Bold ☐
Hour marks markers	Radius 14.4 · Length 1.7 · Width 0.42 ☐
Minutes ticks	Radius 14.4 · Count 60 · Length 1 · Width 0.16 · Skip where markers above ☐
Hour hand	Shape sword · Length 7.6 · Width 1.5 · Tail 1.6 · Tip length 2.4 ☐
Minute hand	Shape sword · Length 12.6 · Width 1.2 · Tail 1.8 · Tip length 3 ☐
Seconds hand	Shape baton · Length 13.2 · Width 0.14 · Tail 3.4 ☐
Cap	Ø 1.5 ☐
Preset	presets/field.json · Dial 30 mm ☒



12.2

Diver

After the 1950s dive watch: fat dots, bars at 6 and 9, a triangle at 12, the date at 3.



Look for
Three markers rings share the hours: dots on **Every 3, skip those instead**; bars on Count 4 with the first position set to none; one triangle. The date takes 3 by itself.

Settings that differ from a new ring

Triangle markers	Radius 13.2 · Count 1 · Shape wedge · Length 3.3 · Width 3.5 · Inner width × 0 · Round outer 0.25 ☐
Bars 6 · 9 markers	Radius 13.2 · Count 4 · Length 3.3 · Width 1.5 · Round outer 0.2 · Round inner 0.2 · 12 o'clock none ☐
Dots markers	Radius 13 · Every 3 · Skip those instead · Shape dot · Ø 2.3 ☐
Depth numerals	Radius 5.2 · Count 1 · Start 180° · Labels Custom · Custom 200 m · 660 ft · Placement on path · Size 0.95 · Font Helvetica Neue Medium ☐
Minutes ticks	Radius 14.3 · Count 60 · Length 0.9 · Width 0.16 ☐
Date window	Position 3 · Radius 10.7 · Width 2.7 · Height 2 · Corner 0.15 · Frame 0 · Print clearance 0.3 · Shows 24 · Text 68 %
Hour hand	Shape pencil · Length 7.4 · Width 2 · Tail 1.6 · Tip length 1.8 ☐
Minute hand	Shape pencil · Length 12.4 · Width 1.5 · Tail 1.8 · Tip length 2 ☐
Seconds hand	Shape lollipop · Length 13 · Width 0.18 · Tail 3.4 · Head · ring · disc 1.25 · at 70 % ☐
Cap	Ø 1.6 ☐
Preset	<i>presets/diver.json</i> · Dial 29 mm ☒



12.3

Dress watch

After the Japanese dress watches of the 1960s: plain bars, a double bar at 12, dauphine hands, a framed date.



Look for
The double bar is the markers ring's 12 o'clock style. Both lines of text are numerals rings with Count 1, set on path at 180°.

Settings that differ from a new ring

Hours markers	Radius 13.9 · Length 2.9 · Width 0.8 · 12 o'clock double · Gap 0.3 ●
Minutes ticks	Radius 14.7 · Count 60 · Length 0.8 · Width 0.09 · Skip where markers above ●
Line 1 numerals	Radius 5.6 · Count 1 · Start 180° · Labels Custom · Custom AUTOMATIC · Placement on path · Size 0.95 · Font Helvetica Neue Medium ●
Line 2 numerals	Radius 7 · Count 1 · Start 180° · Labels Custom · Custom HI-BEAT 36000 · Placement on path · Size 0.7 · Font Helvetica Neue Medium ●
Date window	Position 3 · Radius 10.9 · Width 2.7 · Height 2 · Corner 0.1 · Frame 0.22 · Print clearance 0.25 · Shows 8 · Text 70 %
Hour hand	Shape dauphine · Length 8.4 · Width 1.6 · Tail 1.4 · at 20 % ●
Minute hand	Shape dauphine · Length 13 · Width 1.25 · Tail 1.6 · at 15 % ●
Seconds hand	Shape baton · Length 13.9 · Width 0.12 · Tail 3.6 ●
Cap	Ø 1.1 ●
Preset	presets/dress.json · Dial 30.4 mm ○



12.4

Pilot, type B

After the 1940s observer's watch: minutes outside, hours on an inner ring, a triangle and two dots at 12.



Look for

The minutes are **Numbers**: first 5, step 5, two digits, over a 300° span that leaves 12 free for the triangle. The two dots are a ticks ring with Count 2 on a 22° span.

Settings that differ from a new ring

Minutes numerals	Radius 12.6 · Count 11 · Start 30° · Span 300° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Size 2.5 · Font DIN Alternate Bold ○
Triangle markers	Radius 14 · Count 1 · Shape wedge · Length 2.6 · Width 2.6 · Inner width × 0 ○
Two dots ticks	Radius 13.6 · Count 2 · Start -11° · Span 22° · Shape dot · Ø 0.7 ○
Five minutes markers	Radius 16.4 · Length 2.2 · Width 0.5 ○
Minute track ticks	Radius 16.4 · Count 60 · Length 1.3 · Width 0.2 · Skip where markers above ○
Hours numerals	Radius 6.6 · Size 1.7 · Font DIN Alternate Bold ○
Hour circle band	Radius 8.6 · Band width 0.14 ○
Hour hand	Shape sword · Length 7.4 · Width 2 · Tail 1.4 · Tip length 2.6 ○
Minute hand	Shape sword · Length 14.6 · Width 1.5 · Tail 1.8 · Tip length 3.6 ○
Seconds hand	Shape baton · Length 15.4 · Width 0.16 · Tail 4.2 · Counterweight Ø 1.1 ○
Cap	Ø 1.7 ○
Preset	<i>presets/pilot.json</i> · Dial 34 mm ●



California

Two numerals rings on one radius, each with a **Custom** list that leaves the other's positions empty — so Roman and Arabic can use different fonts and placements.

Roman numerals	Radius 9.6 · Labels Custom · Custom , I, II, , , , , , , X, XI · Placement radial · Size 3.3 · Font Bodoni 72 Book ●
Arabic numerals	Radius 9.6 · Labels Custom · Custom , , , , , 4, 5, , , 7, 8, , , · Placement upright · Size 3.2 · Font DIN Alternate Bold ●
Bars + triangle markers	Radius 11.4 · Count 4 · Length 3.4 · Width 1 · 12 o'clock triangle · Scale × 1.25 ●
Minutes ticks	Radius 14.2 · Count 60 · Length 1 · Width 0.14 ●
Outer rail band	Radius 14.2 · Band width 0.12 ●
Inner rail band	Radius 13.32 · Band width 0.12 ●
Hour hand	Shape pencil · Length 7.2 · Width 1.5 · Tail 1.4 · Tip length 1.6 ●
Minute hand	Shape pencil · Length 11.8 · Width 1.2 · Tail 1.6 · Tip length 1.8 ●
Seconds hand	Shape baton · Length 12.6 · Width 0.13 · Tail 3.2 · Counterweight Ø 0.9 ●
Cap	Ø 1.4 ●
Preset	<i>presets/california.json</i> · Dial 30 mm ●



12.6

Station clock

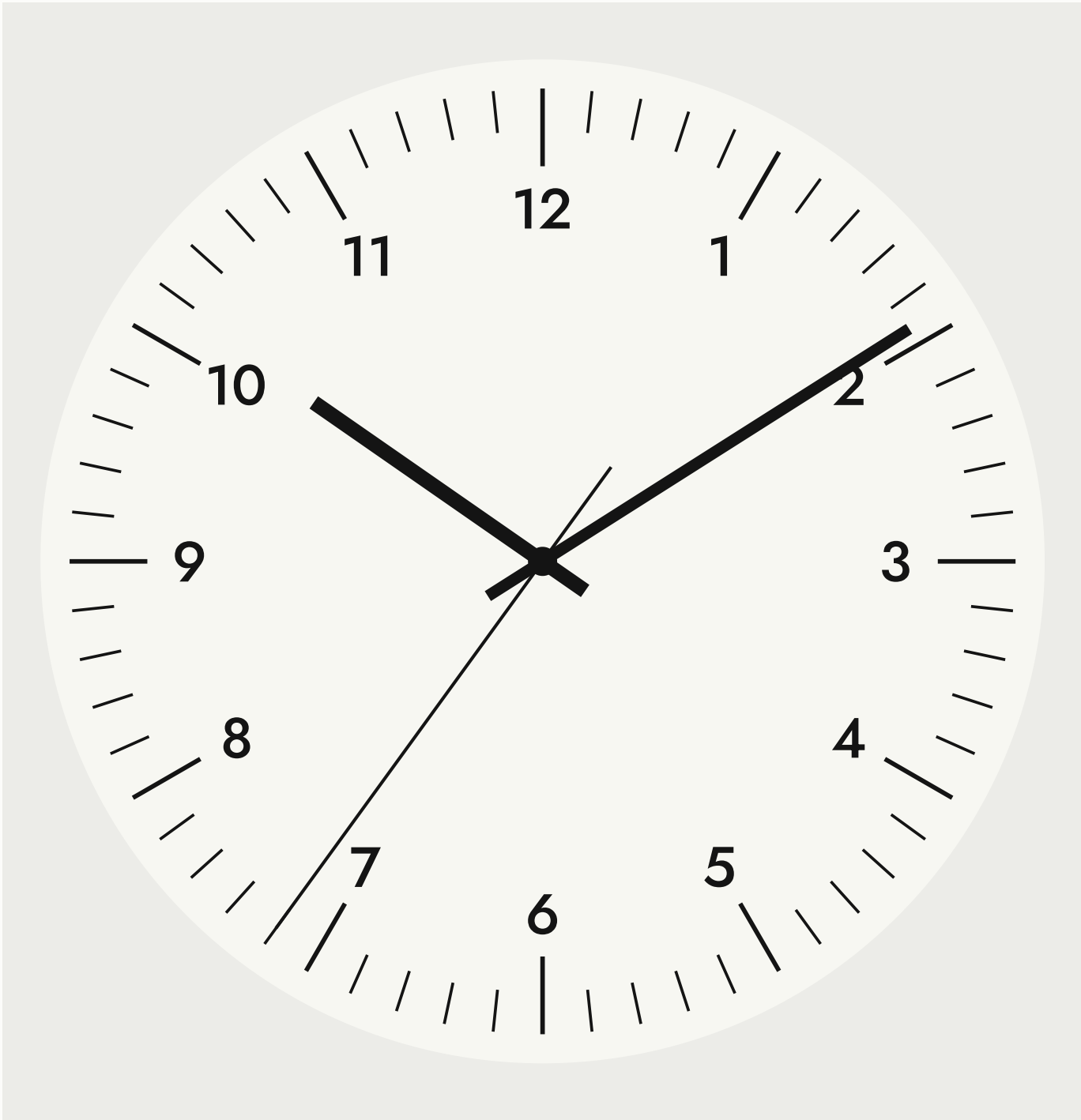
After the Swiss railway clock of 1944: heavy bars, and a red seconds hand like a dispatcher's baton.



Look for
Two rings. Everything is in the proportions: bars 3.7×1.3 , ticks 1.3×0.45 , and a lollipop with its disc at 100 % of the length.

Settings that differ from a new ring

Hours markers	Radius 14.3 · Length 3.7 · Width 1.3 ●
Minutes ticks	Radius 14.3 · Count 60 · Length 1.3 · Width 0.45 · Skip where markers above ●
Hour hand	Shape baton · Length 8.8 · Width 1.7 · Tail 2.4 ●
Minute hand	Shape baton · Length 13.6 · Width 1.3 · Tail 2.8 ●
Seconds hand	Shape lollipop · Length 9.8 · Width 0.34 · Tail 3.6 · Head · ring · disc 2.7 · at 100 % ●
Cap	Ø 1 ●
Preset	<i>presets/railway.json</i> · Dial 30 mm ○



12.7

Bauhaus

After the German wall-clock school of the 1950s–60s: hairline marks, small numerals, thin hands.



Look for
Line weights carry it: 0.14 for the hours, 0.09 for the minutes, hands under half a millimetre.

Settings that differ from a new ring

Hours numerals	Radius 10.9 · Size 1.75 · Font Futura Medium ●
Hour lines markers	Radius 14.6 · Length 2.4 · Width 0.14 ●
Minutes ticks	Radius 14.6 · Count 60 · Length 1.3 · Width 0.09 · Skip where markers above ●
Hour hand	Shape baton · Length 8.6 · Width 0.46 · Tail 1.6 ●
Minute hand	Shape baton · Length 13.4 · Width 0.36 · Tail 2 ●
Seconds hand	Shape baton · Length 14 · Width 0.1 · Tail 3.6 ●
Cap	Ø 0.9 ●
Preset	presets/bauhaus.json · Dial 31 mm ○

Sector dial

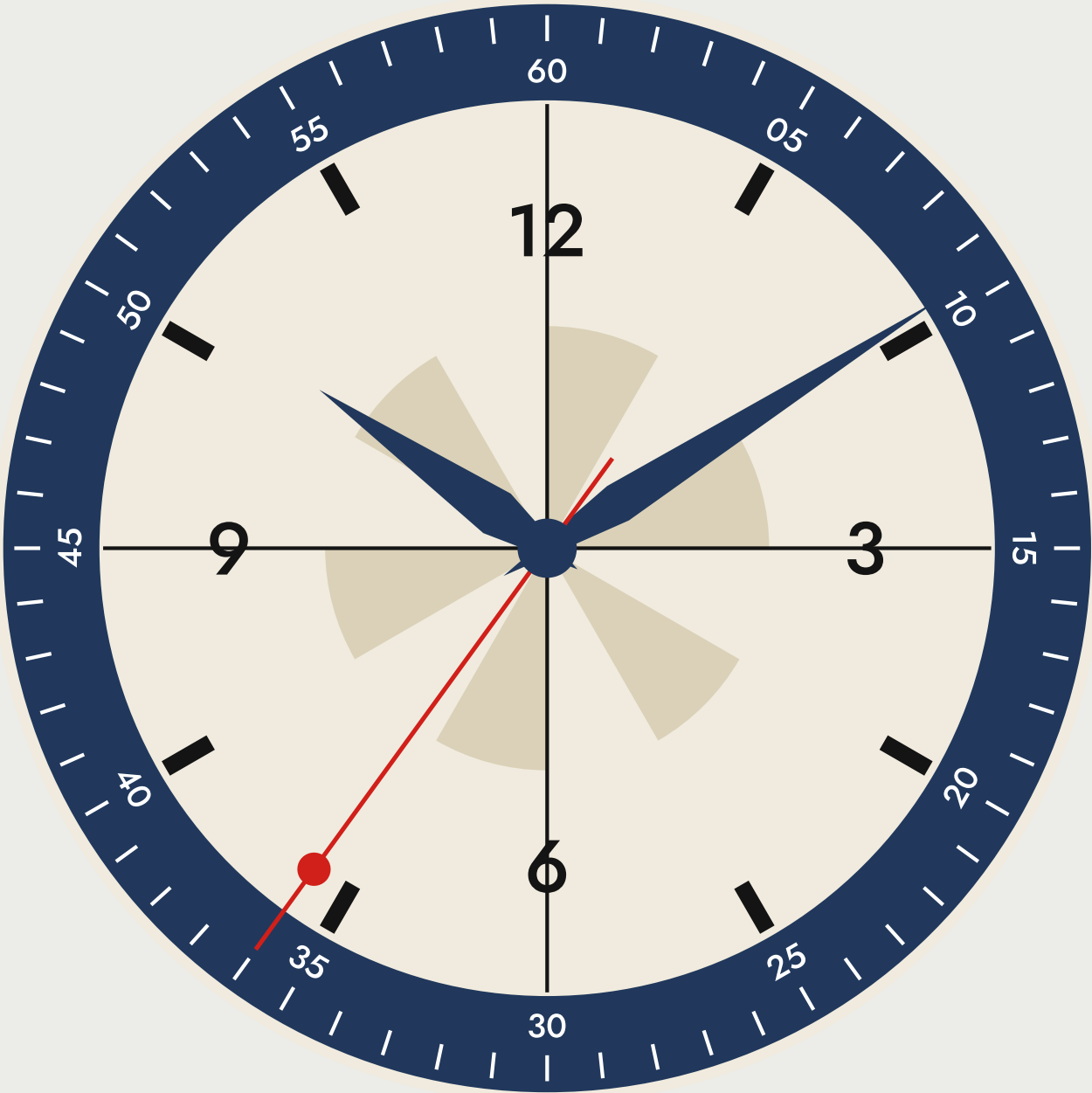
After the 1930s scientific dials: a chapter ring, a crosshair, zones of colour.



Look for
Bands do the colour: a chapter ring 2.6 wide, and
pies made by Band width = Radius with Every 2. The
crosshair is a ticks ring, Count 4, Length 12.

Settings that differ from a new ring

Hours numerals	Radius 8.6 · Every 3 · Size 2 · Font Futura Medium ●
Hour bars markers	Radius 11.9 · Every 3 · Skip those instead · Length 1.4 · Width 0.45 ●
Minutes ticks	Radius 14.4 · Count 60 · Length 0.7 · Width 0.1 ○
Minute numerals numerals	Radius 12.9 · Start 30° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Placement radial, auto-flip · Size 0.9 · Font Futura Medium ○
Chapter band	Radius 14.7 · Band width 2.6 ●
Crosshair ticks	Radius 12 · Count 4 · Length 12 · Width 0.1 ●
Sectors band	Radius 6 · Count 12 · Every 2 · Band width 6 ○
Hour hand	Shape dauphine · Length 7.5 · Width 1.3 · Tail 1 · at 22 % ●
Minute hand	Shape dauphine · Length 12.6 · Width 1.1 · Tail 1.4 · at 18 % ●
Seconds hand	Shape lollipop · Length 13.4 · Width 0.12 · Tail 3 · Head · ring · disc 0.9 · at 80 % ●
Cap	Ø 1.6 ●
Preset	presets/sector.json · Dial 30 mm ○





12.9

Gauge

A 270° scale with a red zone: positions over a partial span mark both ends.



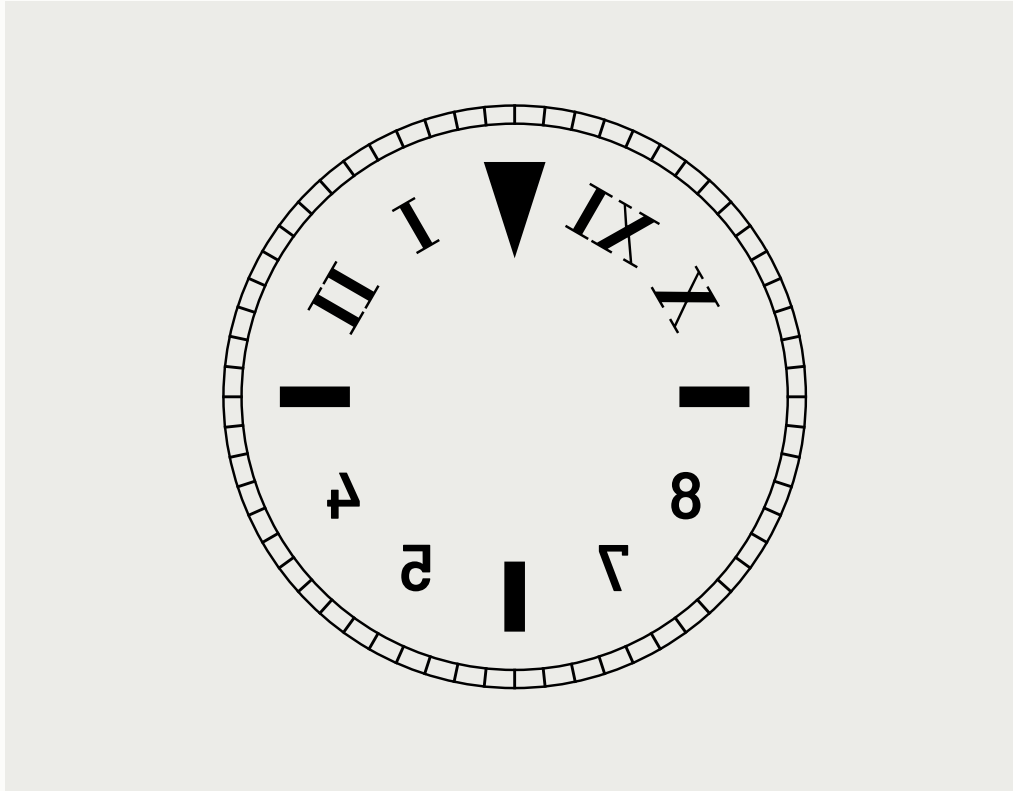
Look for
Start -135°, Span 270°. A partial span marks both ends, so Count 10 gives 0 to 90. The red zone **knocks out** everything above it with a 0.12 clearance.

Settings that differ from a new ring

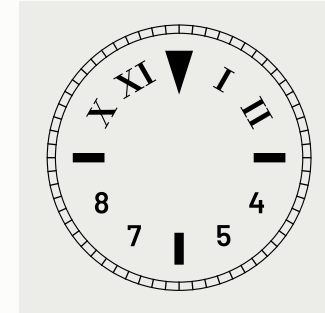
Scale numerals	Radius 11.3 · Count 10 · Start -135° · Span 270° · Labels Numbers · First number 0 · Step 10 · Placement radial, auto-flip · Size 1.5 · Font Helvetica Neue Medium ●
Scale marks markers	Radius 14.3 · Count 10 · Start -135° · Span 270° · Shape wedge · Length 1.6 · Width 0.34 · Inner width × 0.3 ●
Scale ticks ticks	Radius 14.3 · Count 46 · Start -135° · Span 270° · Length 0.8 · Width 0.1 · Skip where markers above ●
Rail band	Radius 14.5 · Count 1 · Start -135° · Span 270° · Band width 0.12 ●
Red zone band	Radius 14.5 · Count 1 · Start 108° · Span 27° · Band width 0.9 · Knock out everything above · Clearance 0.12 ●
Minute hand	Shape arrow · Length 12.6 · Width 0.4 · Tail 2.6 · Tip length 2.2 · Head · ring · disc 1.5 ●
Cap	Ø 1.2 ●
Preset	<i>presets/gauge.json</i> · Dial 30 mm ○

Files in and out

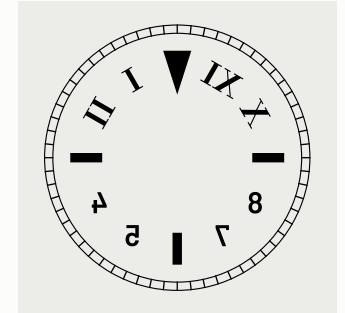
Vectors at true size, a one-colour mask for production, and your own shapes coming in.



design
PDF · SVG · PNG



production
one colour, no plate, no hands



production + mirror
for toner transfer

Export

☒ Production: mono print mask
☒ Mirror: toner transfer
☐ Live text in PDF · SVG (fonts n...)

PNG dpi

Clip seconds

Frames per s



PDF · SVG · PNG · MP4 · GIF

True size.

A 30 mm dial opens 30 mm wide. PNG at your dpi: 30 mm at 600 dpi is 709 pixels. MP4 and GIF run the hands for Clip seconds.



Production · Mirror
One colour, flipped.

Production drops the plate and the hands and prints every ring in one colour: a mask. Mirror flips it for toner transfer, where the print lands face down.



Watch

This is that mask, mirrored. Check its thinnest line at 100 % before you etch: a 0.09 mm tick is two pixels at 600 dpi.



Live text

Off by default.

Keeps numerals editable. The fonts are then needed where the file is opened, and on-path text becomes one object per letter.



In: SVG · PDF · AI

Your own shapes.

Draw pointing up; the artboard's centre is the anchor. Black is ink, white cuts holes. Outline text first.

Reference

Constructions to build on, every setting with its range, and a way in when you know what you want but not its name.



Track

Railroad

Ticks, Count 60. Two bands 0.12 wide: at the ticks' Radius, and at Radius – Length + 0.12. Page 19.



Track

Fifths of a second

Ticks Count 60 above ticks Count 300. The 300 Skip where: all marks above.



Hours

24-hour ring

Numerals, Labels Custom: 24, 13, 14 ... 23, at a smaller Radius and Size. Page 15.



Hours

Two fonts, one circle

Two numerals rings on one Radius; each Custom list leaves the other's positions empty. Page 19.



Minutes

05 to 60

Numerals, Labels Numbers, First number 5, Step 5, Digits 2, Start 30°. Page 22.



Text

A wordmark

Numerals, Count 1, Start 0° or 180°, Labels Custom, Placement on path. Page 17.



Colour

A gap round numerals

A band below the numerals; Knock out: markers & numerals above; Clearance 0.3.



Colour

Sector dial

A band, Count 12, Every 2, Band width equal to Radius. Page 22.



When something is wrong

Read the log, compare the previews, send both.

- 1 Open Export and read the log. At launch it says “preview: canvas” and “all controls connected”.
- 2 If the preview looks wrong, switch Canvas to PDF and compare.
- 3 Send a screenshot and the log. If the window doesn't build, run `tools/dial_probe.py` and send its report.



A ring is gone

Look at its checkbox in the list, its Every and Offset, and whether it skips positions a ring above takes.



No numeral at 3

The date window took it: markers and numerals leave its position empty.



Where things live

One repository.

Code, words, rules github.com/duztman/dial-tool · *docs/*

Presets *docs/guide/presets*

This book *docs/guide/pages*, built by *tools/make_guide.py*



Slow

The log should say “shape combining: skia-pathops”. If not, install it: page 2.



Start over

Export → Settings → Defaults. The last session is kept in *~/Library/Application Support/DialTool*.

Every ring setting

Printed from the tool's own rows, so it cannot drift. Starts at: a markers ring added to a 30 mm dial.

A ring	From · to	Step	Starts at	Shown for
Kind	ticks · markers · numerals · band			all
Colour	a colour		●	all
Outer radius	0 to 25	0.05	13.40	ticks · markers · band
Centre radius	0 to 25	0.05	13.40	numerals
Positions				
Count	1 to 360	1	12	all
Start °	−180 to 180	0.5	0.0	all
Span °	1 to 360	1	360	all
Every	1 to 60	1	1	all
Offset	0 to 59	1	0	all
Skip those instead	on · off		off	all
Give way to rings above				
Skip where	nothing · markers above · markers & numerals above · all marks above			all
Knock out	nothing · markers above · markers & numerals above · everything above			all
Clearance	0 to 2	0.01	0.25	all
Look				
Shape	bar · wedge · dot · file			ticks · markers
File	SVG · PDF · AI			ticks · markers (file)
Size ×	0.05 to 10	0.01	1.00	ticks · markers (file)
Length	0.05 to 10	0.05	2.80	ticks · markers (bar · wedge)
Width · Ø	0.02 to 5	0.01	0.70	ticks · markers (bar · wedge · dot)
Inner width ×	0 to 2	0.01	0.35	ticks · markers (wedge)
Round outer	0 to 2	0.01	0.00	ticks · markers (bar · wedge)
Round inner	0 to 2	0.01	0.00	ticks · markers (bar · wedge)
Hollow wall	0 to 1.5	0.01	0.00	ticks · markers

A ring · 2	From · to	Step	Starts at	Shown for
12 o'clock · first position				
Style	same · double · triangle · none			markers
Scale ×	0.5 to 2.5	0.05	1.00	markers
Gap	0 to 3	0.05	0.50	markers
Labels	Arabic · Roman · III · Roman · IV · Eastern Arabic · Chinese · Words · Custom · Numbers			numerals
Custom	text			numerals
First number	−1000 to 1000	1	0	numerals
Step	−100 to 100	1	5	numerals
Digits	0 to 4	1	0	numerals
Placement	upright · radial · radial, auto-flip · on path			numerals
Size	0.3 to 8	0.05	2.40	numerals
Band width	0.02 to 25	0.05	1.00	band
Fill % of step	1 to 100	1	100	band
Nudge one numeral				
Nudge	per numeral: Radius ± · Angle ±° · Rotate ° · Size ×			numerals
Type				
Family	every font installed on the Mac			numerals
Style	the family's styles			numerals
Instance	a variable font's named instances			numerals
Language	default · tr · nl · de · pl · ro · ca · ar · fa · zh-Hans · ja			numerals
Tracking	−200 to 800	5	0	numerals
Features	the font's OpenType features			numerals
Axes	one slider per variable axis			numerals

Every other setting

Dial, hands, date and export. From · to is where the slider stops; a typed value can go past it.

Hands	From · to	Step	Starts at	Shown for
Show hands	on · off		on	
Motion	sweep · quartz · 1/s · 6 beats/s · 8 beats/s			
Cap Ø	0 to 4	0.05	1.60	
Cap colour	a colour			
Hand				
Show this hand	on · off		on	all
Shape	baton · pencil · sword · dauphine · leaf · arrow · syringe · breguet · lollipop · file			all
File	SVG · PDF · AI			file
Fit file to length	on · off		on	file
Size x	0.05 to 10	0.01	1.00	file
Length	0.5 to 30	0.05	10.00	all
Width	0.02 to 4	0.01	1.00	all but file
Width x	0.1 to 5	0.01	1.00	file
Tail	0 to 10	0.05	1.00	all
Tip length	0 to 10	0.05	1.00	pencil · sword · arrow · syringe · breguet
Head width	0 to 6	0.05	1.20	arrow · syringe
Ring · disc Ø	0 to 6	0.05	1.20	breguet · lollipop
Widest at %	0 to 100	1	75	dauphine
Barrel from %	0 to 100	1	75	syringe
Ring · disc at %	0 to 100	1	75	breguet · lollipop
Counterweight Ø	0 to 4	0.05	0.00	all
Hollow wall	0 to 1	0.01	0.00	all
Colour	a colour			all

Dial	From · to	Step	Starts at
Diameter	15 to 50	0.1	30.0
Page margin	0 to 10	0.5	3.0
Colour			
Plate	a colour		
Backdrop	a colour		
Date	From · to	Step	Starts at
Date window	on · off		off
Position	3 · 4:30 · 6 · 9 · 12		
Radius	2 to 18	0.05	10.50
Width	0.5 to 6	0.05	2.60
Height	0.5 to 5	0.05	2.00
Corner	0 to 2	0.05	0.20
Frame	0 to 0.6	0.01	0.15
Print clearance	0 to 1	0.01	0.25
Print colour	a colour		
Shows	text		17
Text % of height	20 to 100	1	62
Export	From · to	Step	Starts at
Production: mono print mask	on · off		off
Mirror: toner transfer	on · off		off
Live text in PDF · SVG (fonts ne...	on · off		off
PNG dpi	72 to 2400	1	600
Clip seconds	1 to 60	1	4
Frames per s	6 to 60	1	30
Export	PDF · SVG · PNG · MP4 · GIF		
Settings	Save... · Load... · Defaults		

Find it

You know what you want on the dial. This says what the tool calls it, and where this book shows it.



Tracks and marks

a minute track	ticks, Count 60	6
a railroad track	two hairline bands	25
fifths of a second	Count 300, skipping	25
marks only at 3 · 6 · 9 · 12	Every 3	6
all but the cardinals	Skip those instead	6
a scale on an arc	Start · Span	23
a triangle at 12	12 o'clock: triangle	7
a double bar at 12	12 o'clock: double	7
lume dots	Shape dot	7
outlined markers	Hollow wall	7
rounded bars	Round outer · inner	7
my own marker	Shape file	24
ticks that stop at markers	Skip where	8
a gap round the numerals	Knock out	8
a crosshair	ticks, Count 4, long	22



Numerals and colour

Roman, IIII or IV	Labels	10
a 24-hour ring	Labels Custom	15
05 to 60	Labels Numbers	18
compass points, words	Labels Custom	10
text on a curve	Placement on path	17
two fonts on one circle	two Custom lists	19
upright or radial	Placement	10
one numeral looks heavy	Nudge: Size ×	11
a variable font's weight	Axes	11
Turkish İ ı	Language	11
wider spacing	Tracking	11
a chapter ring	band + rings above	9
pie sectors	Band width = Radius	9
a red zone	band, Start · Span	23
a hairline circle	Band width 0.12	9



Hands, date, files

another hand shape	Shape	12
a lollipop seconds hand	Ring · disc Ø	20
a skeleton hand	Hollow wall	12
my own hand	Shape file	24
ticking or sweeping	Motion	12
another time shown	Time field	3
the date at 4:30	Position	13
no frame round the date	Frame 0	13
a mask for etching	Production	24
flipped for toner transfer	Mirror	24
a true-size PDF	Export → PDF	24
a 600 dpi PNG	PNG dpi	24
a moving clip	MP4 · GIF	24
keep or reopen a dial	Save... · Load...	2
begin again	Defaults	25